

PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Software Requirements Specification

Online Bookstore

Version 1.0 – Draft

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Revision History

Name	Date	Reason for Changes	Version
M1	October 2026	Initial draft of the Software Requirements Specification	1.0

1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the requirements for the **Online Bookstore** system, Version 1.0.

The purpose of this document is to specify the functional, non-functional, security, interface, and other requirements of the Online Bookstore.

The system provides a web-based platform through which customers can browse and search for books, manage their shopping cart, place orders, and view their order history. Sellers can add book listings to the platform.

This SRS serves as a common reference for the project team during architecture, implementation, testing, and documentation.

1.2 Scope

The Online Bookstore is a web-based e-commerce application for buying and selling books online.

The scope of the current system includes:

- User registration.
- User login and authentication.
- Browsing available books.
- Searching for books.
- Adding book listings by sellers.
- Adding books to a shopping cart.
- Removing books from a shopping cart.
- Placing orders.
- Viewing order history.
- Role-based access control.
- Protection of user data.

The system is intended to support two primary user roles:

- **Customer** – a user who browses and purchases books.
- **Seller** – a user who adds book listings to the platform.

Features not specified in this SRS are not considered part of the current system scope unless they are subsequently approved and documented by the project team.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification
FR	Functional Requirement
NFR	Non-Functional Requirement
SEC	Security Requirement
UI	User Interface
API	Application Programming Interface
RBAC	Role-Based Access Control
Customer	Registered user who purchases books through the platform
Seller	Registered user who can add book listings
Book Listing	Information representing a book available on the platform
Cart	Collection of books selected by a customer for purchase
Order	Record of books selected by a customer for purchase
Authentication	Process of verifying the identity of a user
Authorization	Process of determining whether a user is allowed to perform an operation
Django	Python-based web framework used to develop the Online Bookstore
ORM	Object-Relational Mapping

1.4 References

1. PES University, Department of Computer Science and Engineering, *Software Requirements Specification Template*.

2. PES University, Department of Computer Science and Engineering, *Online Bookstore Mini-Project Guidelines*.

1.5 Overview

This document is organized into the following major sections:

- **Section 1 – Introduction:** Describes the purpose, scope, terminology, references, and organization of the SRS.
- **Section 2 – Overall Description:** Describes the product perspective, major functions, user characteristics, operating environment, constraints, and assumptions.
- **Section 3 – Specific Requirements:** Defines the external interfaces, functional requirements, non-functional requirements, security requirements, quality attributes, and business rules.
- **Appendix A – Glossary:** Defines important terminology.
- **Appendix B – Field Layouts:** Describes important data fields used by the system.
- **Appendix C – Requirement Traceability Matrix:** Provides traceability between requirements and architecture, design, code, and test cases.

2 Overall Description

2.1 Product Perspective

The Online Bookstore is a new, self-contained web-based application developed as an e-commerce platform for books.

The system consists of a web-based user interface, application logic, authentication and authorization components, and a database.

The application will be implemented using the **Python Django web framework**.

The major logical components of the system are:

- User Authentication and Authorization
- Book Management

- Search and Browsing
- Shopping Cart Management
- Order Management
- Database Management

The detailed architecture and component interactions will be documented in the Software Architecture and Design document.

2.2 Product Functions

The major functions of the Online Bookstore are summarized below.

User Authentication

- Register a new user.
- Authenticate registered users.

Book Management

- Browse available books.
- Search for books.
- Allow sellers to add book listings.

Shopping Cart Management

- Add books to the shopping cart.
- Remove books from the shopping cart.

Order Management

- Place an order.
- View previously placed orders.

Security

- Enforce role-based access.
- Protect user data.

2.3 User Characteristics

The system has two primary user roles.

Customer

A Customer is a registered user who uses the platform to browse and purchase books.

A Customer shall be able to:

- Register for an account.
- Log in to the system.
- Browse books.
- Search for books.
- Add books to the cart.
- Remove books from the cart.
- Place orders.
- View order history.

Seller

A Seller is a registered user who can add book listings to the platform.

A Seller shall be able to:

- Register for an account.
- Log in to the system.
- Browse books.
- Search for books.
- Add book listings.

Role-Based Access

The system shall restrict protected functionality according to the user's assigned role.

Customers shall be permitted to perform customer-related purchasing operations, while sellers shall be permitted to add book listings.

Users shall not be permitted to access functionality that is not authorized for their assigned role.

2.4 Constraints

The following constraints apply to the project:

- The application shall be implemented using the Python Django web framework.
- The system shall operate as a web-based application.
- The system shall use a database for storing application data.
- User access shall be controlled according to assigned roles.
- User data shall be protected from unauthorized access and disclosure.
- The implementation shall remain consistent with the requirements specified in this SRS.
- Functional and non-functional requirements shall be testable.

2.5 Assumptions and Dependencies

Assumptions

- Users have access to a compatible web browser.
- Users have network connectivity when accessing the system.
- Users provide valid information during registration.
- Sellers provide valid information while adding book listings.
- Users have the appropriate role required to perform protected operations.

Dependencies

- The system depends on the Django framework.
- The system depends on a database for storing user, book, cart, and order information.
- The system depends on network connectivity for client-server communication.
- The system depends on the selected deployment environment for hosting the web application.

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The system shall provide web-based interfaces for:

- User registration.
- User login.
- Book browsing.
- Book search.
- Seller book addition.
- Shopping cart management.
- Order placement.
- Order history.

The user interface shall provide clear navigation and appropriate feedback for successful and unsuccessful operations.

3.1.2 Software Interfaces

The system shall interact with the following software components:

- Web browser.
- Django application.
- Database management system.
- Django ORM.

The exact database technology and deployment configuration will be finalized during implementation.

3.1.3 Communications Interfaces

The Online Bookstore shall use standard web-based communication between the client and the Django application.

The system shall support communication required for:

- User authentication.
- Book-related operations.
- Cart operations.
- Order operations.
- Data retrieval and submission.

HTTP/HTTPS shall be used for web communication as supported by the deployment environment.

3.1.4 Analysis Models

The following analysis models shall be included in the final SRS where applicable:

- UML Use Case Diagram.
- Entity-Relationship Diagram.
- Other relevant analysis models.

The UML Use Case Diagram will be provided by M4.

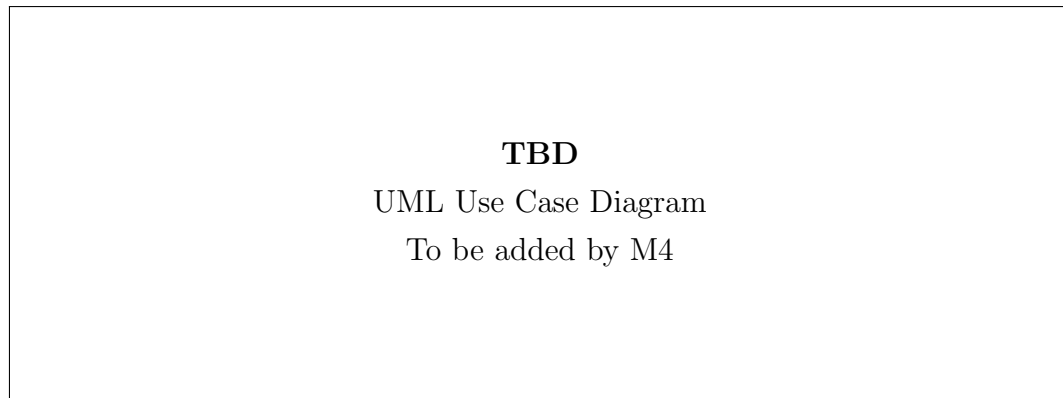


Figure 1: Online Bookstore Use Case Diagram

3.2 Functional Requirements

The functional requirements are organized according to the major system features.

3.2.1 User Authentication

Description and Priority

This feature provides registration and login functionality for users.

Priority: High

Stimulus/Response Sequences

1. The user selects the registration option.
2. The system displays the registration form.
3. The user provides the required registration information.
4. The system validates the submitted information.
5. The system creates the user account when the information is valid.
6. The user provides authentication credentials for login.
7. The system validates the credentials.
8. The system grants access when the credentials are valid.

Functional Requirements

FR-AUTH-01:

The system shall allow a new user to register for an account by providing the required registration information.

FR-AUTH-02:

The system shall allow a registered user to log in using valid authentication credentials.

3.2.2 Book Management

Description and Priority

This feature allows users to browse and search for books and allows authenticated sellers to add book listings.

Priority: High

Stimulus/Response Sequences

1. The user accesses the book section.
2. The system displays available books.
3. The user enters supported search information.
4. The system retrieves matching books.
5. An authenticated seller selects the option to add a book.
6. The seller enters the required book information.
7. The system validates and stores the book listing.

Functional Requirements

FR-BOOK-01:

The system shall allow users to browse the books available on the platform.

FR-BOOK-02:

The system shall allow users to search for books using supported book information.

FR-BOOK-03:

The system shall allow an authenticated seller to add a book listing by providing the required book information.

3.2.3 Shopping Cart Management

Description and Priority

This feature allows authenticated customers to manage books selected for purchase.

Priority: High

Stimulus/Response Sequences

1. The customer selects a book.
2. The customer selects the option to add the book to the cart.
3. The system adds the selected book to the customer's cart.
4. The customer selects a book already present in the cart.
5. The customer selects the remove option.
6. The system removes the selected book from the cart.

Functional Requirements

FR-CART-01:

The system shall allow an authenticated customer to add a book to the shopping cart.

FR-CART-02:

The system shall allow an authenticated customer to remove a book from the shopping cart.

3.2.4 Order Management

Description and Priority

This feature allows authenticated customers to place orders and view previously placed orders.

Priority: High

Stimulus/Response Sequences

1. The customer selects books and adds them to the shopping cart.
2. The customer selects the option to place an order.
3. The system validates the cart information.
4. The system creates the order.
5. The customer accesses the order history.
6. The system displays the customer's previously placed orders.

Functional Requirements

FR-ORDER-01:

The system shall allow an authenticated customer to place an order containing books selected in the shopping cart.

FR-ORDER-02:

The system shall allow an authenticated customer to view their previously placed orders.

3.3 Non-Functional Requirements

3.3.1 NFR-01 – Performance

The system shall display book search results within **3 seconds** under normal operating conditions.

3.3.2 NFR-02 – Usability

The system shall allow a user to complete the major operations—login, book search, cart operation, and order placement—through clear and consistent navigation.

3.3.3 NFR-03 – Reliability

The system shall maintain order and cart data consistency after every successfully completed operation.

3.3.4 NFR-04 – Availability

The system shall be available for normal operations for at least **99% of scheduled operating time**, excluding planned maintenance.

3.3.5 NFR-05 – Compatibility

The system shall support the latest stable versions of:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

3.3.6 NFR-06 – Maintainability

The system shall use modular software components so that changes to one major functional area do not require modification of unrelated functional areas.

3.3.7 NFR-07 – Testability

Each functional requirement shall be independently testable through defined test cases.

3.4 Design Constraints

The system shall satisfy the following design constraints:

- The application shall be implemented using Python Django.
- The system shall be implemented as a web-based application.
- The system shall use a database for persistent application data.
- The system shall enforce role-based access.
- The system shall protect user data from unauthorized access.
- The system shall maintain modularity between major functional components.

3.5 Security Requirements

3.5.1 SEC-01 – Role-Based Access

The system shall enforce role-based access so that users can access only functions permitted for their assigned roles.

Priority: High

3.5.2 SEC-02 – Protection of User Data

The system shall protect user data from unauthorized access and disclosure.

Priority: High

3.5.3 Security Objectives

The security objectives shall be finalized by M4.

Security Objective 1: TBD

Security Objective 2: TBD

3.6 Software Quality Attributes

The following software quality attributes are specified through the non-functional requirements.

Attribute	Requirement	Description
Performance	NFR-01	Search results shall be displayed within 3 seconds under normal operating conditions.
Usability	NFR-02	Major operations shall be accessible through clear and consistent navigation.
Reliability	NFR-03	Cart and order data shall remain consistent after successful operations.
Availability	NFR-04	The system shall provide at least 99% scheduled availability excluding planned maintenance.

Compatibility	NFR-05	The system shall support the latest stable versions of specified browsers.
Maintainability	NFR-06	The system shall use modular software components.
Testability	NFR-07	Functional requirements shall be independently testable.

3.7 Business Rules

The following business rules are derived from the specified system requirements.

- **BR-01:** A user shall register before using functions requiring an authenticated account.
- **BR-02:** Only an authenticated seller with the appropriate role shall be allowed to add book listings.
- **BR-03:** Only an authenticated customer with the appropriate role shall be allowed to perform customer purchasing operations.
- **BR-04:** Users shall access system functionality according to their assigned role.
- **BR-05:** Access restrictions shall be applied consistently to protected system functions.

3.8 Other Requirements

3.8.1 Database Requirements

The system shall maintain persistent data required for:

- User accounts.
- User roles.
- Book listings.
- Shopping carts.
- Orders.

The final database schema will be specified in the architecture and design documentation.

3.8.2 Technology Requirement

The Online Bookstore shall be implemented using the **Python Django web framework**.

Django shall be used for:

- Web application development.
- Application logic.
- User authentication.
- Role and permission management.
- Database interaction through Django ORM.

A Glossary

Term	Meaning
Customer	A registered user who browses and purchases books.
Seller	A registered user who adds book listings to the platform.
Book Listing	Information representing a book available on the platform.
Cart	A collection of books selected by a customer for purchase.
Order	A record of books selected by a customer for purchase.
Authentication	The process of verifying a user's identity.
Authorization	The process of determining whether a user is permitted to perform an operation.
Role	A category assigned to a user that determines permitted operations.
RBAC	Role-Based Access Control.
Django	A Python-based web framework used to implement the Online Bookstore.
ORM	Object-Relational Mapping used for interaction between application objects and database data.

B Field Layouts

The following field layouts provide the initial data requirements. Final field definitions shall be synchronized with the database design.

B.1 User Registration

Field	Data Type	Mandatory	Description
Name	String	Yes	Name of the user.
Email	String	Yes	Email address used for the user account.
Password	String	Yes	Authentication credential for the user account.
Role	String	Yes	Assigned user role such as Customer or Seller.

B.2 Book Listing

Field	Data Type	Mandatory	Description
Book Title	String	Yes	Title of the book.
Author	String	Yes	Author of the book.
Price	Numeric	Yes	Price of the book.
Description	String	No	Description of the book.

B.3 Shopping Cart

Field	Data Type	Mandatory	Description
Customer ID	Identifier	Yes	Identifies the customer associated with the cart.
Book ID	Identifier	Yes	Identifies the selected book.
Quantity	Integer	Yes	Number of copies selected.

B.4 Order

Field	Data Type	Mandatory	Description
Order ID	Identifier	Yes	Unique identifier for the order.
Customer ID	Identifier	Yes	Identifies the customer who placed the order.
Order Date	Date/Time	Yes	Date and time at which the order was placed.
Total Amount	Numeric	Yes	Total amount associated with the order.
Status	String	Yes	Current status of the order.

C Requirement Traceability Matrix

The Requirement Traceability Matrix maps each requirement to its architecture, design, implementation, and testing references.

No.	Requirement ID	Brief Description	Architecture	Design	Code	Test Case	System Test
1	FR-AUTH-01	User registration	TBD	TBD	TBD	TBD	TBD
2	FR-AUTH-02	User login	TBD	TBD	TBD	TBD	TBD
3	FR-BOOK-01	Browse books	TBD	TBD	TBD	TBD	TBD
4	FR-BOOK-02	Search books	TBD	TBD	TBD	TBD	TBD
5	FR-BOOK-03	Seller adds books	TBD	TBD	TBD	TBD	TBD
6	FR-CART-01	Add book to cart	TBD	TBD	TBD	TBD	TBD
7	FR-CART-02	Remove book from cart	TBD	TBD	TBD	TBD	TBD
8	FR-ORDER-01	Place order	TBD	TBD	TBD	TBD	TBD
9	FR-ORDER-02	View order history	TBD	TBD	TBD	TBD	TBD
10	NFR-01	Performance	TBD	TBD	TBD	TBD	TBD
11	NFR-02	Usability	TBD	TBD	TBD	TBD	TBD
12	NFR-03	Reliability	TBD	TBD	TBD	TBD	TBD
13	NFR-04	Availability	TBD	TBD	TBD	TBD	TBD
14	NFR-05	Compatibility	TBD	TBD	TBD	TBD	TBD
15	NFR-06	Maintainability	TBD	TBD	TBD	TBD	TBD
16	NFR-07	Testability	TBD	TBD	TBD	TBD	TBD
17	SEC-01	Role-based access	TBD	TBD	TBD	TBD	TBD

18	SEC-02	Protect user data	TBD	TBD	TBD	TBD	TBD
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